

Determine linear versus superlinear optimal menu size in binary-state information-selling mechanisms

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

This paper gives a complete solution to the two-state full-incentive-compatibility menu-size question under the standard weak-IC convention. For every integer $N \geq 2$, we construct a two-state information-selling instance with three nonredundant actions and a finite distribution on N buyer types such that the minimum cardinality of a revenue-optimal menu is exactly N . The proof bounds the revenue of every incentive-compatible (IC) and individually rational (IR) mechanism by a sum of virtual-value terms. A direct likelihood-ratio majorization argument identifies the unique maximizing information structure for each term over all experiments, not merely over the proposed one-parameter family. We then construct prices implementing all these maximizers and prove global incentive compatibility. Consequently, there is no finite upper bound on optimal menu size that depends only on the number of actions. In particular, the answer to the question of whether optimal menu size must be $O(m)$, where m denotes the number of actions, is negative.

1 The open problem

This paper addresses an open problem posed by Dirk Bergemann, Yang Cai, Grigoris Velezgas, Mingfei Zhao in *Is Selling Complete Information (Approximately) Optimal?* [1] (Economics and Computation (EC), 2022), Section 4.1, discussion immediately after Theorem 3 (page 1011 in the provided text pagination).. The website catalogue entry is problem 19 (W4226352527_p0) of the [Open Problems in Operations Research](#) collection.

In the binary-state information-selling model with m actions, determine whether the cardinality of a revenue-optimal IC and IR menu must always grow linearly in m , or whether it can exceed linear growth in m .

Equivalently, resolve whether for the original (full IC) problem the menu size of an optimal mechanism is always $O(m)$, or whether there exist environments for which every optimal menu has size $\omega(m)$.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260824_061524_c2503abe, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2 Model and result

There are two states, denoted ω_1 and ω_2 . A buyer of type $\theta \in [0, 1]$ has prior

$$\Pr(\omega_1) = \theta, \quad \Pr(\omega_2) = 1 - \theta.$$

The finite action set is $\mathcal{A}$, and U_{ra} denotes the payoff from action $a \in \mathcal{A}$ in state ω_r .

An experiment E with a finite (but arbitrarily large) signal set S is a row-stochastic matrix

$$E = (\pi_{rs})_{r \in \{1,2\}, s \in S}, \quad \pi_{rs} \geq 0, \quad \sum_{s \in S} \pi_{rs} = 1 \quad (r = 1, 2).$$

After observing a signal, the buyer may choose any action. Thus the gross value of E to type θ is

$$V_\theta(E) = \sum_{s \in S} \max_{a \in \mathcal{A}} \{ \theta \pi_{1s} U_{1a} + (1 - \theta) \pi_{2s} U_{2a} \}. \quad (1)$$

This maximization over actions is part of the definition of value, so all incentive constraints below are full incentive constraints and do not presume obedience to signal recommendations.

Without purchasing information, type θ obtains

$$B_\theta = \max_{a \in \mathcal{A}} \{ \theta U_{1a} + (1 - \theta) U_{2a} \}.$$

We write

$$W_\theta(E) = V_\theta(E) - B_\theta$$

for the incremental value of the experiment.

A menu entry consists of an experiment and an ex ante price. If type θ_i is assigned entry (X_i, p_i) , its incremental utility is

$$u_i = W_{\theta_i}(X_i) - p_i.$$

Individual rationality (IR) requires $u_i \geq 0$. Incentive compatibility (IC) requires

$$W_{\theta_i}(X_i) - p_i \geq W_{\theta_i}(X_j) - p_j \quad \text{for all support types } i, j.$$

We use the standard weak-IC direct-mechanism convention: when several entries maximize a type's utility, the designated truthful entry may be selected. Menu cardinality excludes the exogenous no-purchase option.

Experiments that differ only by relabeling signals or by splitting a signal into several proportional columns are regarded as the same information structure. The lower bound below holds even under this identification.

Theorem 1 (Unboundedness with three actions). *For every integer $N \geq 2$, there is a two-state information-selling environment with exactly three nonredundant actions and a probability distribution supported on N types such that:*

1. *there is a revenue-optimal weakly IC and IR menu with exactly N entries; and*
2. *every revenue-optimal weakly IC and IR menu contains at least N distinct information structures.*

Consequently, already with three actions,

$$\sup_{U, F} \{ \text{minimum cardinality of a revenue-optimal menu} \} = \infty,$$

where the supremum ranges over payoff matrices U and type distributions F . Thus no finite menu-size bound can depend only on the number of actions.

3 The construction

Fix $N \geq 2$. For $i = 1, \dots, N$, define

$$d_i = \frac{1}{3 \cdot 2^{i-1}}, \quad \theta_i = 1 - d_i. \quad (2)$$

In particular,

$$d_{i+1} = \frac{d_i}{2}, \quad \theta_i \geq \frac{2}{3}.$$

Set

$$\delta = \frac{d_N}{16}, \quad \alpha = 1 - \delta, \quad \beta = \frac{1}{4}. \quad (3)$$

Since $N \geq 2$,

$$0 < \delta \leq \frac{1}{96}, \quad \frac{95}{96} \leq \alpha < 1. \quad (4)$$

There are three actions, A , M , and C , with payoff matrix

$$U = \begin{pmatrix} 1 & \alpha & 0 \\ 0 & \beta & 1 \end{pmatrix}, \quad (5)$$

where the columns correspond to A , M , and C , respectively.

For support type θ_i , the expected no-information payoff from M is

$$B_i = \alpha\theta_i + \beta d_i. \quad (6)$$

The payoff from A is θ_i , and the payoff from C is d_i . Because $\delta \leq d_i/16$ and $\theta_i < 1$,

$$B_i - \theta_i = \frac{1}{4}d_i - \delta\theta_i > \frac{1}{4}d_i - \frac{1}{16}d_i = \frac{3}{16}d_i > 0. \quad (7)$$

Moreover, using $\theta_i \geq 2/3$, $d_i \leq 1/3$, and $\alpha \geq 95/96$,

$$B_i - d_i = \alpha\theta_i - \frac{3}{4}d_i \geq \frac{95}{96} \cdot \frac{2}{3} - \frac{1}{4} = \frac{59}{144} > 0. \quad (8)$$

Thus M is the unique optimal no-information action of every support type. Actions A and C are uniquely optimal at the degenerate beliefs on ω_1 and ω_2 , respectively. Hence all three actions are nonredundant.

For $q \in [0, 1]$, consider the one-sided disclosure experiment

$$E(q) = \begin{pmatrix} 1 & 0 & 0 \\ 1-q & 0 & q \end{pmatrix}. \quad (9)$$

The third signal reveals ω_2 . The first signal pools all of ω_1 with the remaining state- ω_2 probability. The middle column is a zero-probability column and may be omitted.

Define

$$s_i = 1 - \frac{\delta\theta_i}{\beta d_i} = 1 - \frac{4\delta\theta_i}{d_i}. \quad (10)$$

Lemma 1 (Value of one-sided disclosure). *For every support type i and every $q \in [0, 1]$,*

$$w_i(q) = W_{\theta_i}(E(q)) = \frac{3}{4}d_i q + \frac{1}{4}d_i(q - s_i)_+, \quad (11)$$

where $x_+ = \max\{x, 0\}$. Moreover,

$$s_i - s_{i+1} = \frac{4\delta}{d_i} > 0 \quad (i = 1, \dots, N-1), \quad s_N = 1 - \frac{\theta_N}{4} > \frac{3}{4}. \quad (12)$$

Proof. On the signal revealing ω_2 , action C is uniquely optimal, and its contribution to gross value is $d_i q$.

On the pooled signal, the unnormalized payoffs from A , M , and C are, respectively,

$$\theta_i, \quad \alpha\theta_i + \beta d_i(1 - q), \quad d_i(1 - q).$$

Action C is never optimal on this signal because

$$d_i(1 - q) \leq d_i < \theta_i.$$

The difference between the payoffs from M and A is

$$\alpha\theta_i + \beta d_i(1 - q) - \theta_i = \beta d_i(1 - q) - \delta\theta_i.$$

Thus M is optimal for $q \leq s_i$, while A is optimal for $q \geq s_i$. At $q = s_i$, the two actions tie.

If $q \leq s_i$, subtracting the no-information payoff B_i gives

$$W_{\theta_i}(E(q)) = d_i q + \alpha\theta_i + \beta d_i(1 - q) - B_i = (1 - \beta)d_i q = \frac{3}{4}d_i q.$$

If $q \geq s_i$, then

$$W_{\theta_i}(E(q)) = d_i q + \theta_i - B_i = d_i q - \beta d_i + \delta\theta_i.$$

Since $\beta d_i s_i = \beta d_i - \delta\theta_i$, the two cases combine to give (11).

Finally,

$$\frac{\theta_i}{d_i} = \frac{1}{d_i} - 1, \quad \frac{\theta_{i+1}}{d_{i+1}} = \frac{2}{d_i} - 1,$$

so

$$s_i - s_{i+1} = 4\delta \left(\frac{\theta_{i+1}}{d_{i+1}} - \frac{\theta_i}{d_i} \right) = \frac{4\delta}{d_i} > 0.$$

Because $\delta = d_N/16$,

$$s_N = 1 - \frac{4(d_N/16)\theta_N}{d_N} = 1 - \frac{\theta_N}{4} > \frac{3}{4}. \quad \square$$

Set

$$q_1 = 1, \quad q_i = s_{i-1} \quad (i = 2, \dots, N). \quad (13)$$

Lemma 1 implies

$$1 = q_1 > q_2 > \dots > q_N > \frac{3}{4}. \quad (14)$$

The slopes of w_i , away from its single breakpoint, are $3d_i/4$ and d_i . Hence, for every adjacent pair,

$$\min w'_i = \frac{3}{4}d_i = \frac{3}{2}d_{i+1} > d_{i+1} = \max w'_{i+1}.$$

By continuity and piecewise linearity, it follows (including across the breakpoints) that

$$\Delta_i(q) = w_i(q) - w_{i+1}(q) \quad \text{is strictly increasing on } [0, 1]. \quad (15)$$

Since $w_i(0) = 0$, we also have $\Delta_i(0) = 0$.

4 The type distribution and a universal revenue bound

Let

$$R = 2 - \delta, \quad C_0 = 0, \quad C_i = R^{i-N} \quad (i = 1, \dots, N), \quad (16)$$

and assign probability

$$\mu_i = C_i - C_{i-1} \quad (17)$$

to type θ_i . Since $R > 1$, all μ_i are positive, and

$$\sum_{i=1}^N \mu_i = C_N = 1.$$

For brevity, write

$$W_i(E) = W_{\theta_i}(E).$$

Lemma 2 (Revenue identity and upper bound). *For any IC and IR direct mechanism $(X_i, p_i)_{i=1}^N$, define*

$$u_i = W_i(X_i) - p_i$$

and, for $i = 1, \dots, N-1$, define the adjacent downward-IC slack

$$e_i = u_i - u_{i+1} - (W_i(X_{i+1}) - W_{i+1}(X_{i+1})). \quad (18)$$

Then $e_i \geq 0$, and

$$\begin{aligned} \text{Rev} &= C_1 W_1(X_1) + \sum_{i=2}^N [C_i W_i(X_i) - C_{i-1} W_{i-1}(X_i)] \\ &\quad - u_N - \sum_{i=1}^{N-1} C_i e_i. \end{aligned} \quad (19)$$

Consequently,

$$\text{Rev} \leq C_1 W_1(X_1) + \sum_{i=2}^N [C_i W_i(X_i) - C_{i-1} W_{i-1}(X_i)]. \quad (20)$$

Equality holds in (20) if and only if

$$u_N = 0 \quad \text{and} \quad e_i = 0 \quad (i = 1, \dots, N-1).$$

Proof. The IC constraint preventing type i from selecting type $i+1$'s entry is

$$u_i \geq W_i(X_{i+1}) - p_{i+1}.$$

Since

$$p_{i+1} = W_{i+1}(X_{i+1}) - u_{i+1},$$

this constraint is exactly $e_i \geq 0$.

The finite summation identity

$$\sum_{i=1}^N \mu_i u_i = u_N + \sum_{i=1}^{N-1} C_i (u_i - u_{i+1}) \quad (21)$$

follows by substituting $\mu_i = C_i - C_{i-1}$ and collecting the coefficient of each u_i . Therefore

$$\begin{aligned} \text{Rev} &= \sum_{i=1}^N \mu_i p_i \\ &= \sum_{i=1}^N \mu_i W_i(X_i) - u_N - \sum_{i=1}^{N-1} C_i (u_i - u_{i+1}) \\ &= \sum_{i=1}^N \mu_i W_i(X_i) - u_N - \sum_{i=1}^{N-1} C_i [W_i(X_{i+1}) - W_{i+1}(X_{i+1}) + e_i]. \end{aligned}$$

Reindexing the last sum and using $\mu_i + C_{i-1} = C_i$ gives (19). Individual rationality gives $u_N \geq 0$, and all C_i and e_i are nonnegative, proving the bound and its equality conditions. $\square$

Because $C_i = RC_{i-1}$, the i -th term of (20), for $i \geq 2$, equals

$$C_{i-1} [RW_i(E) - W_{i-1}(E)]. \quad (22)$$

The factor C_{i-1} is strictly positive.

5 Unique maximizers of all virtual terms

5.1 The first term

Lemma 3 (First virtual term). *The function $W_1(E)$ is maximized uniquely, up to signal relabeling and splitting of proportional columns, by full revelation $E(1)$.*

Proof. For a signal column (x, y) , where $x = \pi_{1s}$ and $y = \pi_{2s}$, every action satisfies

$$\theta_1 x U_{1a} + d_1 y U_{2a} \leq \theta_1 x + d_1 y,$$

because every payoff in (5) is at most 1. Summing over columns gives

$$V_{\theta_1}(E) \leq \theta_1 + d_1 = 1.$$

Full revelation attains equality.

Action A is the unique action with payoff 1 in state ω_1 , and action C is the unique action with payoff 1 in state ω_2 . If $x > 0$ and $y > 0$, no single action attains the statewise maximum in both states, so the displayed columnwise inequality is strict. Equality therefore requires every nonzero column to satisfy either $x = 0$ or $y = 0$. Such an experiment separates the two states and is full revelation modulo splitting and relabeling. Subtracting the constant B_1 does not change the maximizer. $\square$

5.2 The adjacent virtual terms

Fix $i \in \{2, \dots, N\}$, and write

$$d = d_{i-1}, \quad \theta = 1 - d, \quad d_i = \frac{d}{2}, \quad \theta' = 1 - \frac{d}{2}. \quad (23)$$

Thus $0 < d \leq 1/3$, and

$$\delta \leq \frac{d_i}{16} = \frac{d}{32}.$$

For a column (x, y) with $y > 0$, define its likelihood ratio

$$\ell = \frac{x}{y}.$$

The gross contribution of this column to type j 's value is $yG_j(\ell)$, where

$$G_j(\ell) = \max \{ \theta_j \ell, \alpha \theta_j \ell + \beta d_j, d_j \}. \quad (24)$$

The three expressions correspond to A , M , and C , respectively.

The two switch points for type j are

$$a_j = \frac{(1 - \beta)d_j}{\alpha \theta_j} = \frac{3d_j}{4\alpha \theta_j}, \quad b_j = \frac{\beta d_j}{\delta \theta_j} = \frac{d_j}{4\delta \theta_j}. \quad (25)$$

Action C is optimal below a_j , action M between a_j and b_j , and action A above b_j .

For the adjacent types in (23),

$$\frac{a_i}{a_{i-1}} = \frac{b_i}{b_{i-1}} = \frac{1 - d}{2 - d} < 1.$$

Furthermore,

$$a_{i-1} = \frac{3d}{4\alpha(1 - d)} \leq \frac{36}{95} < 1,$$

while

$$b_i = \frac{d_i}{4\delta \theta'} \geq \frac{4}{\theta'} > 1.$$

Consequently,

$$0 < a_i < a_{i-1} < 1 < b_i < b_{i-1}. \quad (26)$$

Define the per-unit virtual contribution

$$K(\ell) = RG_i(\ell) - G_{i-1}(\ell). \quad (27)$$

Let

$$b = b_{i-1}, \quad K_0 = K(0) = R\frac{d}{2} - d = -\frac{\delta d}{2}. \quad (28)$$

The breakpoint order (26) gives the complete piecewise description

$$K(\ell) = \begin{cases} R\frac{d}{2} - d, & 0 \leq \ell \leq a_i, \\ R\left(\alpha \theta' \ell + \beta \frac{d}{2}\right) - d, & a_i \leq \ell \leq a_{i-1}, \\ R\left(\alpha \theta' \ell + \beta \frac{d}{2}\right) - (\alpha \theta \ell + \beta d), & a_{i-1} \leq \ell \leq b_i, \\ R\theta' \ell - (\alpha \theta \ell + \beta d), & b_i \leq \ell \leq b, \\ (R\theta' - \theta)\ell, & \ell \geq b. \end{cases} \quad (29)$$

Set

$$\begin{aligned}\gamma &= R\theta' - \theta - \frac{K_0}{b} \\ &= 1 - \delta + \frac{\delta d}{2} + 2\delta^2(1 - d),\end{aligned}\tag{30}$$

$$L(\ell) = K_0 + \gamma\ell.\tag{31}$$

Lemma 4 (Likelihood-ratio majorization). *For every $\ell \geq 0$,*

$$K(\ell) \leq K_0 + \gamma\ell,\tag{32}$$

and equality holds if and only if $\ell = 0$ or $\ell = b$.

Proof. Let $D = L - K$. From (29) and the definition of γ ,

$$D(0) = D(b) = 0.\tag{33}$$

Because $K(\ell) = K_0$ on $[0, a_i]$,

$$D(a_i) = \gamma a_i > 0.\tag{34}$$

Direct substitution at a_{i-1} , using $\alpha\theta a_{i-1} = 3d/4$, gives

$$D(a_{i-1}) = a_{i-1} \left(\gamma - \frac{\alpha R}{2} \right).\tag{35}$$

The coefficient is

$$\gamma - \frac{\alpha R}{2} = \frac{\delta(1+d)}{2} + \delta^2 \left(\frac{3}{2} - 2d \right) > 0,\tag{36}$$

because $\delta > 0$ and $d \leq 1/3$. Hence $D(a_{i-1}) > 0$.

On the interval $[b_i, b]$, the fourth case of (29) and $D(b) = 0$ give

$$D(b_i) = (R\theta' - \alpha\theta - \gamma)(b - b_i).\tag{37}$$

Here

$$R\theta' - \alpha\theta - \gamma = \delta(1-d)(1-2\delta) > 0,\tag{38}$$

because $0 < \delta \leq 1/96 < 1/2$. Since $b_i < b$, $D(b_i) > 0$.

The function D is affine on each interval between consecutive points

$$0, a_i, a_{i-1}, b_i, b.$$

Its values at the two endpoints of every such interval are nonnegative, and they are strictly positive except at 0 and b . Therefore

$$D(\ell) > 0 \quad \text{for } 0 < \ell < b.$$

For $\ell > b$, both types choose A , so

$$D'(\ell) = \gamma - (R\theta' - \theta) = -\frac{K_0}{b} = 2\delta^2(1-d) > 0.$$

Together with $D(b) = 0$, this yields $D(\ell) > 0$ for every $\ell > b$. Thus (32) holds, with equality only at 0 and b . $\square$

Proposition 1 (Unique adjacent virtual maximizer). *For each $i = 2, \dots, N$, the function*

$$E \mapsto RW_i(E) - W_{i-1}(E)$$

is uniquely maximized, up to splitting and relabeling identical-likelihood signals, by

$$E(q_i) = E(s_{i-1}).$$

Proof. First consider a column (x, y) with $y > 0$. Multiplying (32) by y gives

$$yK(x/y) \leq K_0y + \gamma x, \quad (39)$$

with equality if and only if $x/y \in \{0, b\}$.

If $y = 0$ and $x > 0$, both types uniquely choose A . The virtual contribution of the column is

$$x(R\theta' - \theta).$$

Since

$$\gamma - (R\theta' - \theta) = -\frac{K_0}{b} = 2\delta^2(1 - d) > 0, \quad (40)$$

such a state- ω_1 -only column has strict slack relative to $\gamma x + K_0y$. A column with $x = y = 0$ is irrelevant.

Applying the homogeneous inequality to every nonzero column of an arbitrary experiment and using the two row-sum constraints gives

$$\begin{aligned} RV_{\theta_i}(E) - V_{\theta_{i-1}}(E) &\leq \gamma \sum_s \pi_{1s} + K_0 \sum_s \pi_{2s} \\ &= \gamma + K_0. \end{aligned} \quad (41)$$

Equality requires that there be no nonzero state- ω_1 -only column and that every column with positive state- ω_2 probability have likelihood ratio either 0 or b . Let Y_b denote the total state- ω_2 mass of the columns with likelihood b , and let Y_0 denote the corresponding total for likelihood 0. Row normalization then gives

$$bY_b = 1, \quad Y_b + Y_0 = 1.$$

Hence

$$Y_b = \frac{1}{b}, \quad Y_0 = 1 - \frac{1}{b}.$$

After merging columns with the same likelihood, the experiment is therefore

$$\begin{pmatrix} 1 & 0 & 0 \\ 1/b & 0 & 1 - 1/b \end{pmatrix} = E \left(1 - \frac{1}{b} \right).$$

Since $b = b_{i-1} = \beta d_{i-1} / (\delta \theta_{i-1})$,

$$1 - \frac{1}{b} = 1 - \frac{\delta \theta_{i-1}}{\beta d_{i-1}} = s_{i-1} = q_i.$$

Conversely, $E(q_i)$ has only likelihoods b and 0, so it attains equality.

Finally,

$$RW_i(E) - W_{i-1}(E) = RV_{\theta_i}(E) - V_{\theta_{i-1}}(E) - (RB_i - B_{i-1}).$$

The term in parentheses is independent of E , so the gross and incremental virtual objectives have exactly the same maximizers. $\square$

Combining Lemma 3 and Proposition 1, every term in the revenue bound has the unique maximizing allocation

$$X_1 = E(1), \quad X_i = E(q_i) \quad (i = 2, \dots, N), \quad (42)$$

modulo redundant splitting and relabeling.

6 Implementation and global incentive compatibility

Assign experiment $E(q_i)$ to type i . Define rents recursively by

$$u_N = 0, \quad u_i = u_{i+1} + w_i(q_{i+1}) - w_{i+1}(q_{i+1}) \quad (i = N-1, \dots, 1). \quad (43)$$

Equivalently,

$$u_i - u_{i+1} = \Delta_i(q_{i+1}).$$

Because Δ_i is strictly increasing, $\Delta_i(0) = 0$, and $q_{i+1} > 0$,

$$u_i - u_{i+1} > 0. \quad (44)$$

It follows that

$$u_1 > u_2 > \dots > u_N = 0,$$

so all IR constraints hold.

Charge type i the price

$$p_i = w_i(q_i) - u_i. \quad (45)$$

For $i = 1, \dots, N-1$, the rent recursion gives

$$p_i = p_{i+1} + w_i(q_i) - w_{i+1}(q_{i+1}). \quad (46)$$

Also,

$$p_N = w_N(q_N) > 0.$$

Every w_i is strictly increasing and $q_i > q_{i+1}$, so

$$p_1 > p_2 > \dots > p_N > 0. \quad (47)$$

It remains to verify all IC constraints, not merely the adjacent ones. Fix $i < j$. Telescoping (43) gives

$$u_i - u_j = \sum_{k=i}^{j-1} \Delta_k(q_{k+1}). \quad (48)$$

For every $k = i, \dots, j-1$, the ordering (14) implies

$$q_j \leq q_{k+1} \leq q_i.$$

Since every Δ_k is increasing,

$$\Delta_k(q_j) \leq \Delta_k(q_{k+1}) \leq \Delta_k(q_i).$$

Summing and telescoping the differences $w_k - w_{k+1}$ yields

$$w_i(q_j) - w_j(q_j) \leq u_i - u_j \leq w_i(q_i) - w_j(q_i). \quad (49)$$

The left inequality is equivalent to

$$u_i \geq w_i(q_j) - p_j,$$

so type i does not prefer type j 's entry. The right inequality is equivalent to

$$u_j \geq w_j(q_i) - p_i,$$

so type j does not prefer type i 's entry. Thus both IC directions hold for every pair of support types. Because $w_i(q)$ was computed using the optimal action following every signal, these are full IC constraints.

The type distribution has finite support. If a direct mechanism is required to specify an outcome for every off-support type, assign each such type any utility-maximizing member of the finite menu or the no-purchase option. This adds no menu entry and creates no new profitable deviation.

For each $i < N$, type i is indifferent between its designated entry and entry $i + 1$, because the corresponding adjacent downward constraint binds. Type N is indifferent between its entry and the no-purchase option. The implementation therefore uses weak IC exactly as stated in the model; equivalently, ties are resolved toward the designated, higher-priced entry.

7 Optimality and the minimum menu size

For the constructed mechanism, $u_N = 0$, and (43) implies

$$e_i = 0 \quad (i = 1, \dots, N - 1)$$

in the notation of Lemma 2. Hence the mechanism attains equality in the universal revenue bound (20).

Its assigned experiment $X_1 = E(1)$ maximizes the first term by Lemma 3. For every $i \geq 2$, its assigned experiment $X_i = E(q_i)$ maximizes the i -th virtual term by Proposition 1. Thus the mechanism attains the sum of the separate maxima of all terms in (20). No mechanism can obtain greater revenue, so the constructed N -entry menu is revenue-optimal.

Now consider any other revenue-optimal mechanism and designate, for each support type, an entry selected under the weak-IC convention. Its revenue is at most the right-hand side of (20), which in turn is at most the sum of the separate virtual-term maxima. The constructed mechanism attains that sum. Therefore every inequality must be an equality.

All coefficients $C_1, \dots, C_{N-1}$ multiplying the individual virtual objectives are strictly positive. Hence every individual virtual term must attain its own maximum. Lemma 3 and Proposition 1 force

$$X_1 = E(1), \quad X_i = E(q_i) \quad (i = 2, \dots, N),$$

up to redundant splitting and relabeling.

These N information structures are pairwise distinct. Full revelation has only likelihood ratios 0 and $+\infty$. For $i \geq 2$, the only nonzero finite likelihood of $E(q_i)$ is

$$\frac{1}{1 - q_i} = b_{i-1}.$$

The values b_{i-1} are distinct because the q_i are pairwise distinct. Signal splitting and relabeling do not change these likelihoods. Thus no single menu entry can supply the forced structures for two different support types. Every optimal menu consequently contains at least N distinct experiments.

The displayed menu has exactly N entries, so the minimum cardinality of an optimal menu in this instance is exactly N . All inequalities used above remain strict in the boundary case $N = 2$; in particular $\delta = 1/96 > 0$, so that case requires no separate exception. This proves Theorem 1.

8 Consequences for a varying number of actions

Corollary 1. *There is a family of two-state instances with m actions for which every optimal menu has $\omega(m)$ entries. More strongly, no finite upper bound on optimal menu size can be a function only of m .*

Proof. Theorem 1 already proves the stronger fixed-action statement: with $m = 3$, the minimum optimal menu size can be arbitrarily large.

If a family indexed by exactly $m \geq 3$ actions is desired, take $N = m^2$, use the construction above, and append $m - 3$ actions, each with payoff vector

$$(-1, -1).$$

Each appended action is strictly dominated and therefore never affects any value calculation, IC constraint, virtual objective, or equality condition. The resulting instance still requires at least

$$N = m^2$$

menu entries. Since $m^2/m = m \rightarrow \infty$, this lower bound is $\omega(m)$. □

9 Conclusion

The result is a complete solution under the standard weak-IC convention. There is no remaining asymptotic gap: optimal menu size is not merely superlinear in the number of actions, but is unbounded even when the number of nonredundant actions is fixed at three. The likelihood-ratio certificate applies to every feasible experiment with an arbitrary number of signal columns, so the conclusion does not rely on restricting attention to the one-sided disclosure family used by the optimal construction.

References

- [1] Dirk Bergemann, Yang Cai, Grigoris Velezgas, Mingfei Zhao. *Is Selling Complete Information (Approximately) Optimal?*. Economics and Computation (EC), 2022. <https://doi.org/10.1145/3490486.3538304>